

1. Use mediator pattern to let colleague_I communicate with colleagueII .Write a Test class to test the Result.

Colleague

```
package experiment21_1;

abstract class Colleague {
    protected Mediator mediator;

    public Colleague(Mediator mediator){
        this.mediator = mediator;
    }

    public abstract void send(String msg);
    public abstract void receive(String msg);
}
```

Colleague_I

```
package experiment21_1;

class Colleague_I extends Colleague{
    public Colleague_I(Mediator mediator){
        super(mediator);
    }

    @Override
    public void send(String msg) {
        System.out.println("Colleague I is sending message: " +
```

```

msg);

    mediator.sendMsg(this, msg);

}

@Override

public void receive(String msg) {

    System.out.println("Colleague II received message: " +
msg);

}

}

```

Colleague_II

```

package experiment21_1;

class Colleague_II extends Colleague{

    public Colleague_II(Mediator mediator){

        super(mediator);

    }

    @Override

    public void send(String msg) {

        System.out.println("Colleague II is sending message: "
+ msg);

```

```

        mediator.sendMsg(this, msg);
    }

    @Override
    public void receive(String msg) {
        System.out.println("Colleague 1 received message: " +
msg);
    }
}

```

ConcreteMediator

```

package experiment21_1;

class ConcreteMediator implements Mediator{
    private Colleague colleague1;
    private Colleague colleague11;

    public void setColleague1(Colleague colleague1){
        this.colleague1 = colleague1;
    }

    public void setColleague11(Colleague colleague11){
        this.colleague11 = colleague11;
    }
}

```

```

@Override

public void sendMsg(Colleague fromcolleague, String msg)
{

    if(fromcolleague == colleaguel){

        colleaguell.receive(msg);

    }else if(fromcolleague == colleaguell){

        colleaguel.receive(msg);

    }

}

}

```

Mediator

```

package experiment21_1;

interface Mediator {

    void sendMsg(Colleague fromcolleague, String msg);

}

```

Test

```

package experiment21_1;

public class Test {

    public static void main(String[] args) {

        ConcreteMediator mediator = new

```

```
ConcreteMediator();
```

```
    Colleague colleague_I = new Colleague_I(mediator);
```

```
    Colleague colleague_II = new Colleague_II(mediator);
```

```
    mediator.setColleagueI(colleague_I);
```

```
    mediator.setColleagueII(colleague_II);
```

```
    colleague_I.send("Hello, ColleagueII!");
```

```
    colleague_II.send("Hello, ColleagueI!");
```

```
}
```

```
}
```

Output

```
D:\JDK\bin\java.exe "-javaagent:D:\idea\IntelliJ IDEA 2021
```

```
Colleague I is sending message: Hello, ColleagueII!
```

```
Colleague I received message: Hello, ColleagueII!
```

```
Colleague II is sending message: Hello, ColleagueI!
```

```
Colleague II received message: Hello, ColleagueI!
```

```
Process finished with exit code 0
```